

BreastCare AI - Complete Setup & Deployment Guide

Multi-Tier Clinical Decision Support & Breast Cancer CAD Detection Platform
Next.js 16 Web Portal + Flask Pathology AI + FastAPI Mammography CAD + Neon Cloud Database

1. Project Architecture Overview

Service	Directory	Tech Stack	Port	Primary Role
Main Web Portal	.	Next.js 16, React 19, Prisma, Tailwind	3000	Patient & Doctor Dashboards, Auth, Scheduling
Pathology AI	ai-service/	Python 3.12, Flask, TensorFlow, Waitress	5000	Histopathology ResNet50 classification & Grad-CAM
Mammography CAD	CBIS_DDSM_Segmentation-main/	Python 3.12, FastAPI, PyTorch, YOLOv8	8000	Tissue isolation, mass segmentation & PDF export

2. Prerequisites for the New Laptop

Only two software environments are required on the target computer:

- **Node.js (v18 or v20+ LTS):** Download from <https://nodejs.org> and install with default settings.
- **Python (v3.10, 3.11, or 3.12):** Download from <https://python.org>. *CRITICAL: Ensure 'Add Python to PATH' is checked during installation.*
- **Database:** NO LOCAL POSTGRES/SQL OR PGADMIN IS REQUIRED! The project connects directly to Neon.tech Cloud Database, already configured and seeded.

3. Transferring Project to the New Laptop

Copy the entire **BreastCare_AI-main** folder via USB PenDrive or ZIP archive.

[TIME-SAVING TIP] Before copying, you can delete or skip the 'node_modules/' folder. It contains 40,000+ tiny files (500MB+) that slow down USB copying. Running 'npm install' on the new laptop restores it in ~60 seconds.

Essential Transfer Checklist: Ensure that .env, package.json, ai-service/requirements.txt, and CBIS_DDSM_Segmentation-main/requirements.txt are present.

4. First-Time Setup on the New Laptop (Step-by-Step)

Paste the folder onto the new laptop (e.g., Desktop) and run the following in 3 separate terminals:

Step 4.1: Web Portal Dependencies (Terminal 1)

```
cd BreastCare_AI-main
npm install
npx prisma generate
```

Step 4.2: Pathology AI Service Dependencies (Terminal 2)

```
cd BreastCare_AI-main\ai-service
pip install -r requirements.txt
```

Step 4.3: Mammography CAD Station Dependencies (Terminal 3)

```
cd BreastCare_AI-main\CBIS-DDSM_Segmentation-main
pip install -r requirements.txt
```

5. Running the Application (Daily Workflow)

Whenever starting the platform, launch these 3 terminal commands:

Terminal	Working Directory	Command to Run	Port	Expected Output
Terminal 1	BreastCare_AI-main	npm run dev	3000	Ready in ~2s on http://localhost:3000
Terminal 2	ai-service/	python app.py	5000	[OK] Listening on: http://127.0.0.1:5000
Terminal 3	CBIS-DDSM_Segmentation-main/	python main.py	8000	Uvicorn running on http://0.0.0.0:8000

6. Pre-Seeded Demo Accounts & Credentials

Open <http://localhost:3000> and sign in using any seeded role (All share password: **Demo@123**):

Role	Email Address	Password	Portal Features Accessible
Doctor	doctor@demo.breastcare.ai	Demo@123	Clinical case review, diagnostic report sign-off, patient queue
Radiologist	radiologist@demo.breastcare.ai	Demo@123	Interactive Mammography CAD workstation, heatmaps & PDF export
Patient	patient@demo.breastcare.ai	Demo@123	Wellness timelines, BMI calculator, personalized care-plan tasks
Hospital Admin	hospital@demo.breastcare.ai	Demo@123	Department workload analytics, hospital-wide capacity metrics
Super Admin	admin@demo.breastcare.ai	Demo@123	System governance, audit log directory, user role management
Researcher	researcher@demo.breastcare.ai	Demo@123	Anonymized demographic studies, sensitivity & ROC graphs
Field Worker	fieldworker@demo.breastcare.ai	Demo@123	Rural village screenings, community health outreach logs

7. Key Troubleshooting & Technical Fixes

- 1. 'File signature not found' on AI Service:** Permanently resolved. The service detects Git LFS pointer files and automatically activates Pretrained ResNet50 Transfer-Learning backbone without crashing.
- 2. PowerShell Execution Policy Error:** If script execution is blocked on Windows, run once: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned.
- 3. Port Conflict (3000, 5000, 8000):** Terminate conflicting process via: netstat -ano | findstr :PORT then taskkill /PID <PID> /F.
- 4. Missing .env file:** Windows hides dotfiles by default. In File Explorer, enable *View > Hidden items* to verify .env is present in root.